

8.3 The Sound of Music (MUS)

A study of music and recorded sound, focusing on the production of sound by musical instruments and the operation of a CD player:

- synthesised and 'live' sounds
- standing waves in string and wind instruments
- reading a CD by laser.

Waves and photons are used to model the behaviour of light.

There are opportunities for students to develop ICT skills and other skills relating to practical investigation and to communication.

Students discuss environmental issues related to noise.

Students will be assessed on their ability to:	Suggested experiments
28 understand and use the terms <i>amplitude</i> , <i>frequency</i> , <i>period</i> , <i>speed</i> and <i>wavelength</i>	Wave machine or computer simulation of wave properties
29 identify the different regions of the electromagnetic spectrum and describe some of their applications	
30 use the wave equation $v = f\lambda$	
31 recall that a sound wave is a longitudinal wave which can be described in terms of the displacement of molecules	Demonstration using a loudspeaker Demonstration using waves on a long spring
32 use graphs to represent transverse and longitudinal waves, including standing waves	
33 explain and use the concepts of wavefront, coherence, path difference, superposition and phase	Demonstration with ripple tank
34 recognise and use the relationship between phase difference and path difference	

Students will be assessed on their ability to:	Suggested experiments
35 explain what is meant by a <i>standing (stationary) wave</i> , investigate how such a wave is formed, and identify nodes and antinodes	Melde's experiment, sonometer
36 recognise and use the expression for refractive index $\mu_2 = \sin i / \sin r = v_1 / v_2$, determine refractive index for a material in the laboratory, and predict whether total internal reflection will occur at an interface using critical angle	
37 investigate and explain how to measure refractive index	Measure the refractive index of both solids and liquids
38 discuss situations that require the accurate determination of refractive index	
39 investigate and explain what is meant by <i>plane polarised light</i>	Models of structures to investigate stress concentrations
40 investigate and explain how to measure the rotation of the plane of polarisation	
44 recall that, in general, waves are transmitted and reflected at an interface between media	Demonstration using a laser
45 explain how different media affect the transmission/reflection of waves travelling from one medium to another	
63 explain how the behaviour of light can be described in terms of waves and photons	
68 explain atomic line spectra in terms of transitions between discrete energy levels	Demonstration using gas-filled tubes